

1 Algorithm: CBUC

CBUC has two phases.

1. **Offline pruning.** Filter $\mathcal{S}$ to the subset $\mathcal{S}_q$ whose input schemas are type-compatible with the query q and, recursively, with the outputs of already-included tools.
2. **Online sequencing.** Run a budget-aware UCB policy over $\mathcal{S}_q$, stopping when the budget is estimated to be exhausted at confidence $1 - \delta$ or when the estimated marginal utility of every remaining tool falls below a threshold η .

1.1 Offline pruning

Given a type universe $\mathcal{U}$ and schemas $\sigma_i = (\text{In}_i, \text{Out}_i) \in 2^{\mathcal{U}} \times 2^{\mathcal{U}}$, define the query type-closure $\text{cl}(q)$: start with the types mentioned in q , then iteratively add Out_i whenever $\text{In}_i \subseteq \text{cl}(q)$. Set $\mathcal{S}_q = \{s_i : \text{In}_i \subseteq \text{cl}(q)\}$.

Proposition 1 (Completeness of offline pruning). *If a feasible non-adaptive policy $\pi \in \Pi_{\text{non}}$ uses tool s_i , then $s_i \in \mathcal{S}_q$.*

Proof. By induction on call order and the definition of cl . □ □

1.2 Online sequencing

Let $K = |\mathcal{S}_q|$. For each tool $s_i \in \mathcal{S}_q$ maintain empirical mean cost $\hat{c}_i$, mean latency $\hat{\ell}_i$ and mean marginal utility $\hat{u}_i$ from prior invocations (possibly across tasks). At round t with remaining budget B_t and remaining deadline T_t , compute for each feasible tool (satisfying $\hat{c}_i + \beta\sqrt{\log t/N_i} \leq B_t$ and the analogous latency condition) the *upper-confidence utility-per-cost*:

$$\text{UCB}_i(t) = \frac{\hat{u}_i + \alpha\sqrt{\log t/N_i}}{\max(\hat{c}_i - \gamma\sqrt{\log t/N_i}, c_{\min})}, \quad (1)$$

where $\alpha, \beta, \gamma > 0$ are confidence-width parameters and $c_{\min}$ is a lower-bound prior on cost. Select $i_t = \arg \max_i \text{UCB}_i(t)$; return **STOP** if no tool is feasible or the selected tool's posterior marginal utility falls below η . The full procedure is given in Algorithm 2.

Figure 2: CBUC: Cost-Budgeted Upper Confidence

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Require: Tool pool  $\mathcal{S}$ , task  $\tau = (q, B, T, \delta)$ , parameters
 $\alpha, \beta, \gamma, \eta, z$ 
 $\mathcal{S}_q \leftarrow \{s_i : \ln_i \subseteq \text{cl}(q)\}$  /
! /
! offline pruning
 $B' \leftarrow B - z\sigma_c\sqrt{T_{\text{guess}}}$  ;  $T' \leftarrow T - z\sigma_\ell\sqrt{T_{\text{guess}}}$  /
! /
! chance-constraint correction
Initialise  $\hat{c}_i, \hat{\ell}_i, \hat{u}_i, N_i$  for all  $s_i \in \mathcal{S}_q$ 
 $B_t \leftarrow B'$  ;  $T_t \leftarrow T'$  ;  $t \leftarrow 1$  while  $t \leq T_{\max}$  do
 $\mathcal{F} \leftarrow \{i : \hat{c}_i + \beta\sqrt{\log t/N_i} \leq B_t \wedge \hat{\ell}_i + \beta\sqrt{\log t/N_i} \leq T_t\}$  if  $\mathcal{F} = \emptyset$  then
return
STOPendif
 $i_t \leftarrow \arg \max_{i \in \mathcal{F}} \text{UCB}_i(t)$  /
! /
! Eq. 1 if  $\hat{u}_{i_t} < \eta$  then
return
STOPendif
Invoke  $s_{i_t}$ ; observe  $c_{i_t}, \ell_{i_t}, u_{i_t}$ 
Update  $\hat{c}_{i_t}, \hat{\ell}_{i_t}, \hat{u}_{i_t}, N_{i_t}$ 
 $B_t \leftarrow B_t - c_{i_t}$  ;  $T_t \leftarrow T_t - \ell_{i_t}$  ;  $t \leftarrow t + 1$  end while

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1.3 Regret bound

Let $\pi^\dagger \in \Pi_{\text{non}}$ denote the best feasible non-adaptive policy and define

$$\text{REG}(T) = T \cdot U(\pi^\dagger) - \mathbb{E}\left[\sum_{t=1}^T U(h_t)\right].$$

Theorem 2 (Regret of CBUC). *Assume (i) cost, latency and marginal utility are σ -sub-Gaussian and bounded in $[0, M]$, (ii) empirical-Bernstein concentration is applied to $\hat{c}_i, \hat{\ell}_i, \hat{u}_i$, and (iii) the minimum expected cost is*

bounded below by $c_{\min} > 0$. Then CBUC attains

$$\text{REG}(T) = \tilde{\mathcal{O}}(\sqrt{KT \log T}),$$

where $\tilde{\mathcal{O}}$ hides constants depending on M , $c_{\min}$, σ , and on the chance-constraint level δ .

Proof sketch. The result follows from specialising the analysis of Badanidiyuru, Kleinberg & Slivkins [?] for bandits with knapsacks to two resources (budget, deadline) and one reward stream. Two modifications are required. First, the utility-per-cost ratio is handled by the standard reduction to a Lagrangian per-unit-resource reward. Second, the chance-constraint slack is absorbed by tightening the effective budget to $B' = B - \Phi^{-1}(1 - \delta) \sigma \sqrt{T}$, which is a well-defined finite correction under sub-Gaussianity. Applying the BWK regret bound to the modified instance yields the stated rate. A full proof adapting Theorem 4.1 of [?] is given in Appendix ??.

Remark 3. *The bound is against the best non-adaptive competitor; the adaptivity gap (Remark ??) implies the best adaptive policy is at most a constant factor better, so CBUC is also $\tilde{\mathcal{O}}(\sqrt{KT})$ -competitive against the adaptive oracle up to that constant.*

1.4 Complexity

Per round the selection step takes $\mathcal{O}(K \log K)$ time. Over T rounds the total cost is $\mathcal{O}(TK \log K)$ time and $\mathcal{O}(K)$ memory for running statistics. For realistic $K \leq 10^4$ and $T \leq 10^3$ this overhead is negligible compared to any LLM call.